

Pharmacology of DMARDS Part 2: Conventional synthetic DMARDS, Biological DMARDS, and Targeted small molecule synthetic DMARDS

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I am available to groups and individuals for pharmacology help and discussions by appointment.

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Following completion of the preparation materials, students should be able to:

1. List the clinically relevant pharmacokinetic properties of the glucocorticoids and each of the disease modifying antirheumatic drugs (DMARDs) starred (*) on the notes handout.
2. Explain the mechanisms of action and mechanisms of adverse effects of the glucocorticoids, each of the DMARDs classes, and the starred (*) agents using specific examples.
3. Describe the roles of NSAIDs, glucocorticoids, and each class of DMARDs in the treatment of rheumatoid arthritis in relation to their mechanisms and effects using specific examples from among the starred (*) agents.
4. Relate the toxicities of the glucocorticoids and each of the DMARDs to their effects in the various tissues using specific examples.
5. Identify the pretreatment testing and monitoring recommended during therapy in relation to potential adverse effects of DMARDs therapy.
6. Develop a visual aid (concept map, table, or whatever works for you) of the glucocorticoids' and DMARDs' pharmacological effects and risk factors correlated with their potential drug-disease and drug-drug interactions and precautions in special populations (pregnancy and lactation, pediatric and geriatric patients).

Preparation Materials

Required

- ScholarRx Bricks | Practice Questions (questions FOR learning)

Highly relevant optional materials:

- Dr. Goldstein's Word handout | Video Lecture | Guided Reading Questions
- Textbook resources with links are listed on the CPG and on the next slide

SUGGESTIONS:

- Use the resources that work best for you.
- You do not need to study all of them.
- Work through the GUIDED READING QUESTIONS with pen/pencil and paper.

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

Links to resources

ScholarRx Brick > Immunology

- Autoimmune Diseases and Autoinflammatory Disease: <https://exchange.scholarrx.com/collection/immunology>
- Treatment of Immune Disorders: <https://exchange.scholarrx.com/brick/rheumatoid-arthritis>
- Biologic Agents: <https://exchange.scholarrx.com/brick/biologic-agents>

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 36: Nonsteroidal Anti-inflammatory Drugs, Disease Modifying Antirheumatic Drugs, Nonopioid Analgesics & Drugs Used in Gout

The section on Nonsteroidal Anti-inflammatory Drugs (NSAIDs)

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382§ionid=281752931>

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 39: Adrenocorticosteroids & Adrenocortical Antagonists > The Naturally Occurring Glucocorticoids; Cortisol (hydrocortisone) – The sections on Naturally Occurring Glucocorticoids and Synthetic Corticosteroids > Adrenocorticosteroids

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382§ionid=281753522#1204142345>

LWW Heath Library, Premium Basic Sciences: Lippincott's Illustrated Reviews: Pharmacology, 8e, 2023; Chapter 40: Anti-inflammatory, Antipyretic, and Analgesic Agents

<https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=253330080&bookid=3222>

Access Medicine Katzung's Pharmacology: Examination and Board Review, 14e, 2024; Chapter 36: NSAIDs, Acetaminophen & Drugs Used in Rheumatoid Arthritis & Gout

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3461§ionid=285596163>

Introductory Case Vignette

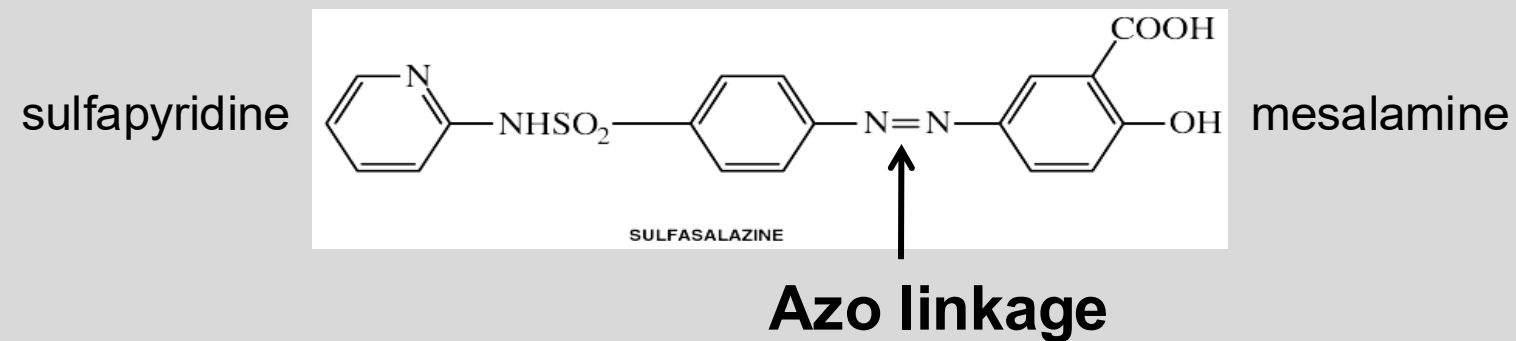
- Justine is a 45-year-old female diagnosed with moderately to severely active RA. The rheumatologist discusses early aggressive therapy with her.
- Justine agrees to oral triple therapy with glucocorticoid bridging.

What medications will Justine be receiving for RA treatment?

What monitoring for adverse effects should be done relative to each drug?

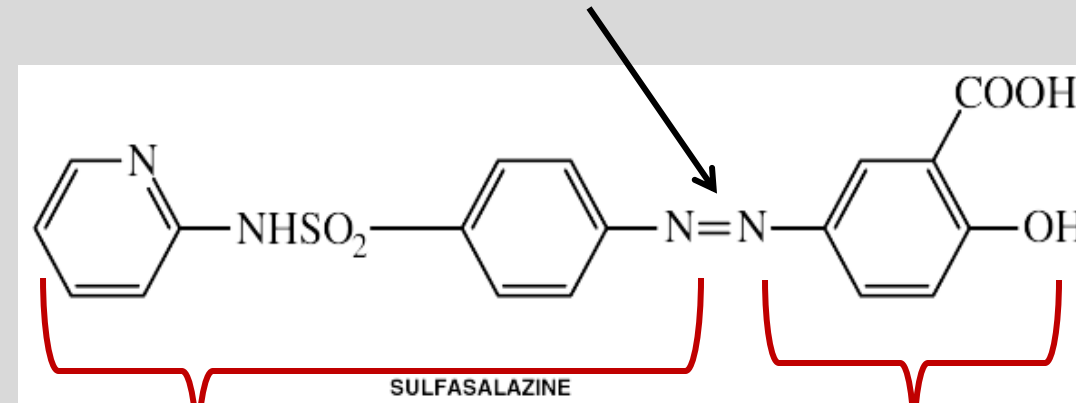
*Sulfasalazine

Mesalamine (5-aminosalicylic acid) is connected to sulfapyridine by an azo linkage.



Sulfasalazine

Enteric bacterial enzyme azoreductase
cleaves azo linkage



sulfapyridine + mesalamine

(Mesalamine is the active component in the treatment of inflammatory bowel disease.)

thought to be active in RA;
metabolites excreted in urine

minimal absorption
excreted in feces

Sulfasalazine Pharmacokinetic Properties

A	Oral sulfasalazine <15% bioavailability Cleaved sulfapyridine bioavailability ~60%; 5-ASA minimal absorption
D	Highly albumin-bound (sulfonamides class effect) Sulfasalazine has high affinity for connective tissue. Sulfapyridine and sulfasalazine: adequate concentration in synovial fluid
M	Sulfapyridine: Hepatic acetylation ➤ Rate of metabolism is dependent on the N-acetyltransferase (NAT2) phenotype (pharmacogenomics)
E	Renal excretion: metabolites and unchanged drug
t_{1/2}	~10 hours in fast acetylators ~15 hours in slow acetylators
	Onset of action in RA: 6 weeks – 3 months

Can displace other
albumin-bound drugs

Slow acetylators may have ↑ risk of toxicity

Sulfasalazine's Mechanism of Action and Efficacy in RA

Mechanism of action: Unknown

- Studies suggest sulfapyridine is the active moiety.

Postulated actions

(based on in vitro and in vivo studies)

- Suppression of T cell responses
- Inhibition of B cell proliferation
- Inhibition of proinflammatory cytokine release by monocytes/macrophages
 - eg IL-1, IL-6, IL-12, TNF α
- ↓ IgA and IgM rheumatoid factor production (in RA patients)

Efficacy in the treatment of RA

- Reduces radiologic disease progression and other therapeutic benefits
 - SSZ monotherapy, and
 - Combination with nonbiological and biological DMARDS
- Combinations may be more effective than single agents utilized alone

Example of common combination:

MTX + sulfasalazine ± hydroxychloroquine

- Appears to be safe during pregnancy
- Sulfasalazine is approved for juvenile idiopathic arthritis (and ulcerative colitis) in children ≥ 6 years old

Sulfasalazine Adverse Effects / Cautions

Common*

- *Headache
- *GI effects
- *Rash
- Oligospermia (reversible)
- Hemolytic anemia

in patients with glucose-6-phosphate dehydrogenase (G6PD) deficiency (?)

Note: Evidence is inconclusive.

Drug interactions:

- Sulfasalazine impairs folic acid absorption.
Folic acid supplementation (1 mg daily) is recommended in all patients and higher folic acid dose in pregnant patients (2 mg daily).
- Additive adverse effects.

Life-threatening toxicities

(occur rarely)

Delayed hypersensitivity reactions

- Stevens-Johnson syndrome; toxic epidermal necrolysis
- Drug reaction with eosinophilia and systemic symptoms (DRESS)
 - Hepatotoxicity
 - Pulmonary toxicity
 - Agranulocytosis; neutropenia
 - Interstitial nephritis

**DISCONTINUE AT 1ST SIGN
OF RASH OR OTHER SERIOUS TOXICITY**

Giant urticarial eruption in patient with drug reaction to sulfa



Source: Usatine RP, Smith MA, Mayeaux EJ Jr, Chumley H, Tysinger J:
The Color Atlas of Family Medicine: www.accessmedicine.com
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Toxic epidermal necrolysis on the face and large areas of desquamation on the leg



Source: Usatine RP, Smith MA, Mayeaux EJ Jr, Chumley H, Tysinger J:
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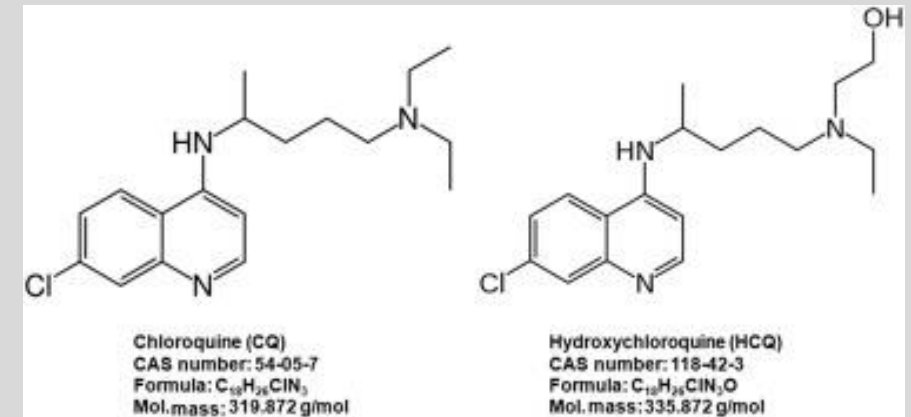
- *Hydroxychloroquine (HCQ)
 - Chloroquine (CQ)

HCQ and CQ are equally efficacious.

HCQ is preferred because it is generally accepted to have a better side effect profile.

GI intolerance is more common with CQ.

HCQ may cause less ocular toxicity.



HCQ (and CQ) Pharmacokinetic Properties

A	Rapid complete absorption from GI tract
D	Moderate plasma protein binding Extensively sequestered in tissues (V_d range up to 800 L/kg): liver, spleen, melanin-containing tissues including eyes, erythrocytes brain, heart, kidneys, leukocytes, and lungs
M	Extensive hepatic metabolism: HCQ minor CYP metabolism to inactive metabolites Chloroquine via CYP2D6 and 3A4 (inhibits CYP2D6) to inactive metabolite
E	Renal excretion: metabolites and unchanged drug
$t_{1/2}$	30-50 DAYS traces of drug can be found in urine for years after therapy
	Onset of action in RA: 6 weeks – 3 months

HCQ / CQ Mechanism of Action / Efficacy in RA

Concentrate inside cells
in acidic vesicles



↑ pH of lysosomal and
endosomal compartments

May suppress T-cell responses

- ↓ leukocyte chemotaxis
- stabilize lysosomal enzymes
- inhibit DNA and RNA synthesis
- trap free radicals

Accumulate in white blood cells

- Stabilize lysosomal membranes
- Inhibit enzyme activity, e.g.
collagenase; proteases that cause
cartilage breakdown

Therapeutic uses

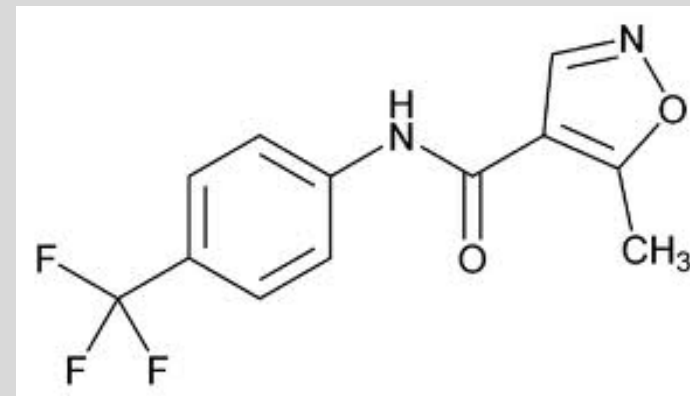
- Monotherapy of mildly active RA
- Combination therapy:
MTX + sulfasalazine + HQ

HCQ / CQ Adverse Effects in RA Treatment

Dose and duration related effects
(believed to be related to drug accumulation in tissues)

Common	• GI upset: nausea, vomiting, cramping, and diarrhea • Headache, lightheadedness	
Serious (very rare)	• skin reactions • psychosis, seizures • myopathy • retinal damage • QT interval prolongation	
Ocular effects	• retinopathy (may be irreversible) • corneal deposits (reversible)	Ophthalmologic exam at baseline and annual monitoring is recommended
Pregnancy Lactation	• considered safe in all trimesters • considered safe in breastfeeding infants	

*Leflunomide

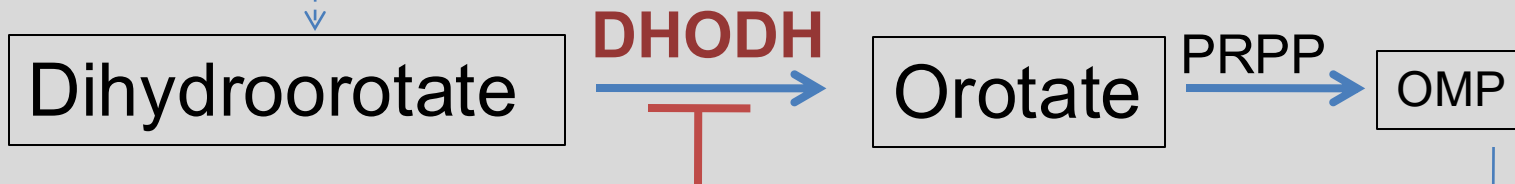


Leflunomide Pharmacokinetic Properties

A	Prodrug ; well absorbed
D	Plasma protein binding >99% of active metabolite (M1 = teriflunomide)
M	Rapid nonenzymatic conversion to active M1= teriflunomide by first-pass metabolism, probably in the intestinal wall and liver ➤ Enterohepatic circulation of teriflunomide, which contributes to the long $t_{1/2}$
E	Feces / urine ➤ Cholestyramine can enhance leflunomide excretion by chelating recirculated leflunomide in the gut, which increases total clearance by approximately 50%. This mechanism is important when leflunomide levels must be rapidly reduced, such as in severe leflunomide toxicity or when a patient wished to become pregnant.
$t_{1/2}$	18 DAYS (mean)
	Onset of action in RA: 4-8 weeks

Teriflunamide: DHODH Competitive Inhibitor → inhibits de novo pyrimidine synthesis

Aspartate + CO₂ + NH₃



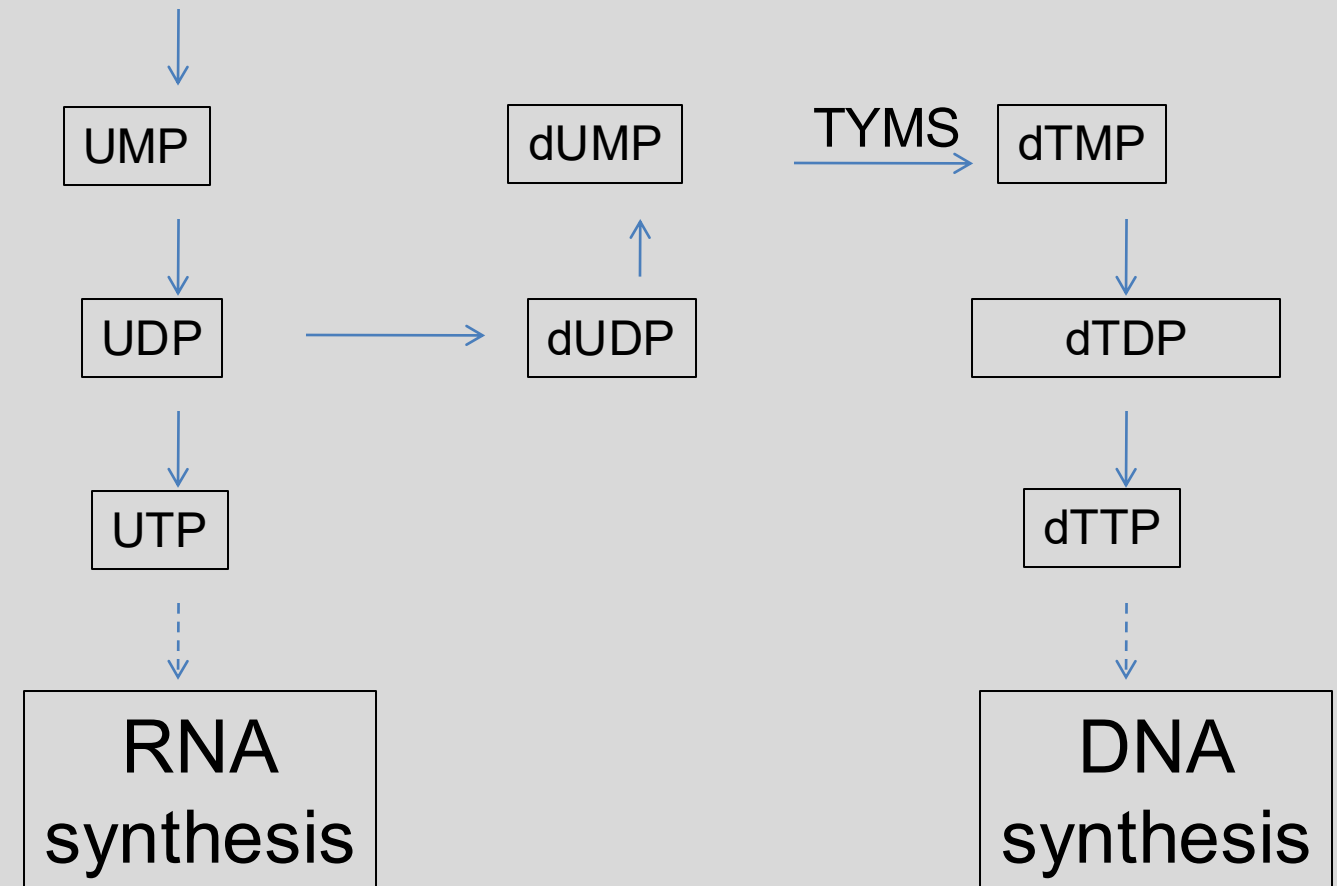
Leflunomide → Teriflunomide

Cell cycle block in G1/S phase

Activates apoptotic pathways

PRPP: 5-phosphoribosyl-1-pyrophosphate
DHODH: dihydroorotate dehydrogenase
TYMS: thymidylate synthase

**antiproliferative and
anti-inflammatory effects**



Leflunomide inhibits T cell proliferation, reduces production of autoantibodies by B cells, blocks NF-κB activation, which blocks proinflammatory effects of this nuclear factor, and decreases IL-2 synthesis and increases IL-10 secretion.

Leflunomide Toxicities

Effect	Comment → Monitor Therapy	
GI toxicity	Diarrhea (common), nausea	
Severe liver injury (US Boxed Warning)	Fulminant liver failure; hepatic necrosis; cirrhosis - Fatalities have occurred - Concurrent hepatotoxic drugs or hepatic disease → ↑risk <i>Risk factors:</i> Higher dose Patients with preexisting chronic or acute liver disease Concomitant drugs that cause liver toxicity, including methotrexate	1. Screen for tuberculosis and pregnancy prior to therapy 2. Monitor CBC, LFTs; and signs/symptoms of severe infection, hepatotoxicity <i>In the event of severe leflunomide toxicity:</i> 3. Cholestyramine to sequester teriflunomide in gut for rapid elimination
Bone marrow suppression	Pancytopenia, agranulocytosis, thrombocytopenia reported <i>Risk factor:</i> Concurrent or recent methotrexate	
Dermatologic	Severe cutaneous adverse reactions SJS/TEN/DRESS	
And	interstitial lung disease; peripheral neuropathy; infection <i>Risk factor:</i> History of methotrexate use	

Leflunomide Special Populations

- **PREGNANCY** (Category X) ***Contraindicated***
 - Leflunomide has been associated with teratogenic and **embryolethal** effects in animal models **at low doses**.
 - **Women of childbearing potential should not receive therapy until pregnancy has been excluded, they have been counseled concerning fetal risk and reliable contraceptive measures have been confirmed.**
- **LACTATION** Excretion in breast milk unknown/not recommended potential for serious adverse reactions in nursing infant
- **PEDIATRICS** Not approved for use in children

Recap: csDMARDs

	SSZ GI effects, headache, rash, folate def. reversible oligospermia hemolysis in G6PD def.	Mechanisms unknown Triple therapy Pregnancy: relatively safe
MTX The “anchor” Initial Tx and combo with all DMARDs	Suppression of immune attack on synovial tissues. Onset: weeks-months	HCQ “weak” DMARD GI, ocular, skin toxicity
Antimetabolites Toxicities: liver, bone marrow, GI, lung, neuro, skin Fetotoxic/Teratogenic	LEF Alternative to MTX CYP2C19 very long $t_{1/2}$ enterohepatic circ.	

Check your knowledge

June is a 28-year-old woman given a diagnosis of moderate RA without features of a poor prognosis. She was recently married and reports that she and her husband are planning to start a family.

Which of the following would be an appropriate initial RA monotherapy for this patient?

- Hydroxychloroquine
- Leflunomide
- Methotrexate
- Sulfasalazine

SJ is a 45-year-old man who presents to the ED with complaints of nausea and vomiting, lack of appetite, right upper quadrant pain, abdominal distention, generalized itching, jaundice, fatigue, and confusion. Medical history is significant for moderate to severe RA treatment with an oral csDMARD. Right upper quadrant tenderness, hepatomegaly, and ascites are noted on PE.

Prothrombin time is prolonged and aminotransferase and bilirubin levels are elevated. The patient is admitted for management of drug-induced acute liver failure. Cholestyramine is included in the treatment to accelerate drug elimination by preventing enterohepatic circulation of the RA drug.

- What is the drug?

Back to Mary's case:

Mary presents to the rheumatologist for a 3-month evaluation of her rheumatoid arthritis treatment.

- In the past month, Mary experienced a disease flare treated with a 3-week course of high-dose prednisone followed by dose tapering and withdrawal of the glucocorticoid. She is tolerating the methotrexate.
- As monotherapy with MTX alone was inadequate for controlling disease activity, Mary is offered oral triple therapy or MTX / bDMARD therapy.
- Mary opts for the biologic agent because therapeutic response is achieved more quickly than with the triple therapy. She refuses the intravenous bDMARD option and agrees to add-on therapy with adalimumab self-injected subcutaneously every other week, which is covered by her insurance plan.

Adalimumab is in what class of drugs (mechanism-based classes)?

What is the intravenous option that Mary preferred not to receive?

What are other therapy options for the treatment of RA resistant to methotrexate monotherapy?

Targeted Immunosuppressive Agents

bDMARDs (parenteral)

Anti-TNF agents

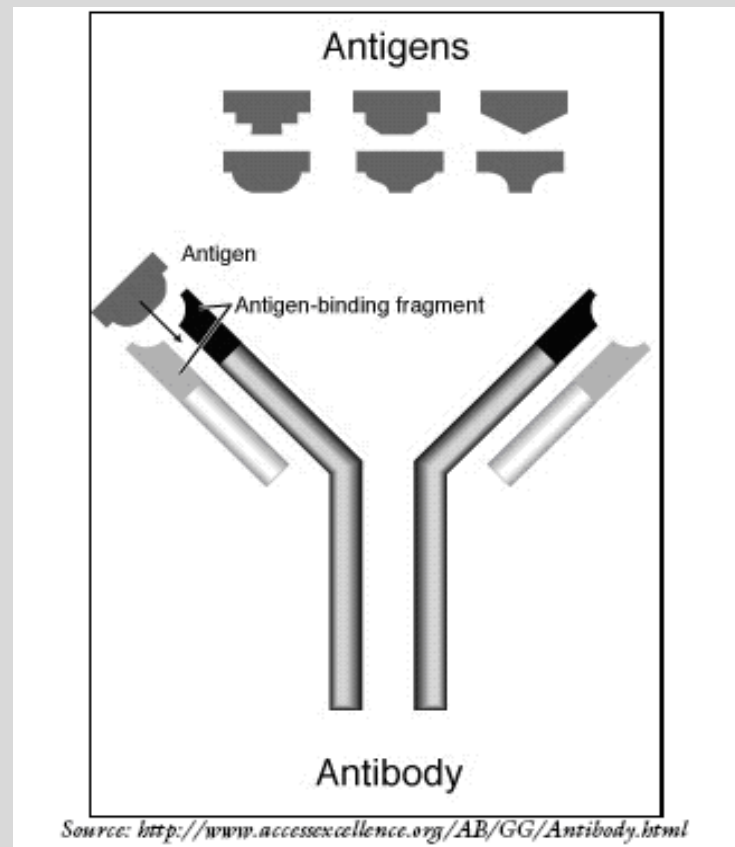
Interleukin-6 receptor antagonists

Modulator of T lymphocytes

Modulator of B lymphocytes

tsDMARDs (oral)

Janus kinase (JAK) Inhibitors



Class Properties of the Targeted Immunosuppressants

Effect	Comment
Route of administration	Parenteral (I.V. or subQ): bDMARDs are polypeptides, which are not absorbed from GI tract Oral: tsDMARDs
Mechanism of action	Target specific mediators of inflammation or receptor proteins on inflammatory cells
Therapeutic use	Combination therapy or monotherapy (agent-specific)
Adverse effects	Infection; lymphoma; myelosuppression; hepatic injury; lupus-like syndrome; hypertension, heart failure; immunological reactions

Targeted Immunosuppressants Class Considerations

Effect	Comment
Immunizations	Admin prior to beginning treatment with immunosuppressant to protect against infection Vaccines given concurrently may have diminished immunologic effect; risk of infection from attenuated live vaccines (theoretical risk, no data)
Screening	Exposure to tuberculosis; HBV, HCV; PML (rituximab)
Monitoring	Response; hypersensitivity rx; malignancy
Pregnancy	Studies not conducted
Lactation	Excretion in breast milk unknown
Pediatrics	See individual drugs

Contraindications related to the effects of the drugs

The drugs cause...	Contraindications: Do not give to patients with...
Immunosuppression	Myelosuppression / Immunosuppressive disease Cancer Concurrently with other immunosuppressants
Infection	Infectious disease
Hepatic injury	Hepatic disease
Hypertension	Hypertensive disease (can exacerbate heart failure)
Allergic reaction	Hypersensitivity reaction to the drug or murine proteins or any other component
Pregnancy	Relative contraindication
Other biologic DMARD	Increased risk of serious infection

Anti-TNF Agents

- * Infliximab
- * Etanercept
- * Adalimumab
 - Golimumab
 - Certolizumab pegol

Monoclonal Antibodies and Fusion Proteins

Structures of TNF- α antagonists
used in rheumatoid arthritis.

C_H, constant heavy chain

C_L, constant light chain

F_C, complex immunoglobulin region

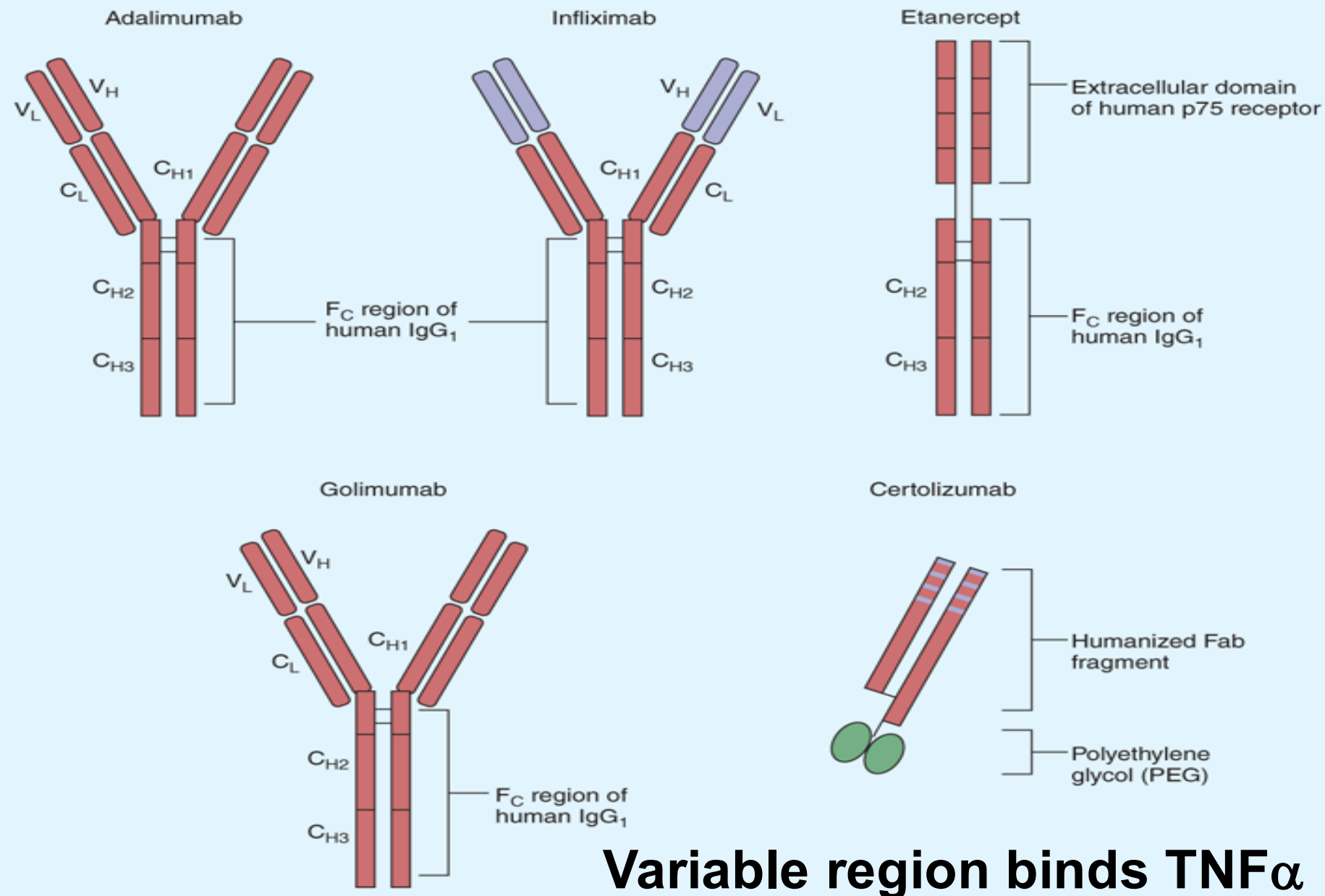
V_H, variable heavy chain

V_L, variable light chain

Red regions, human derived

Blue regions, mouse derived

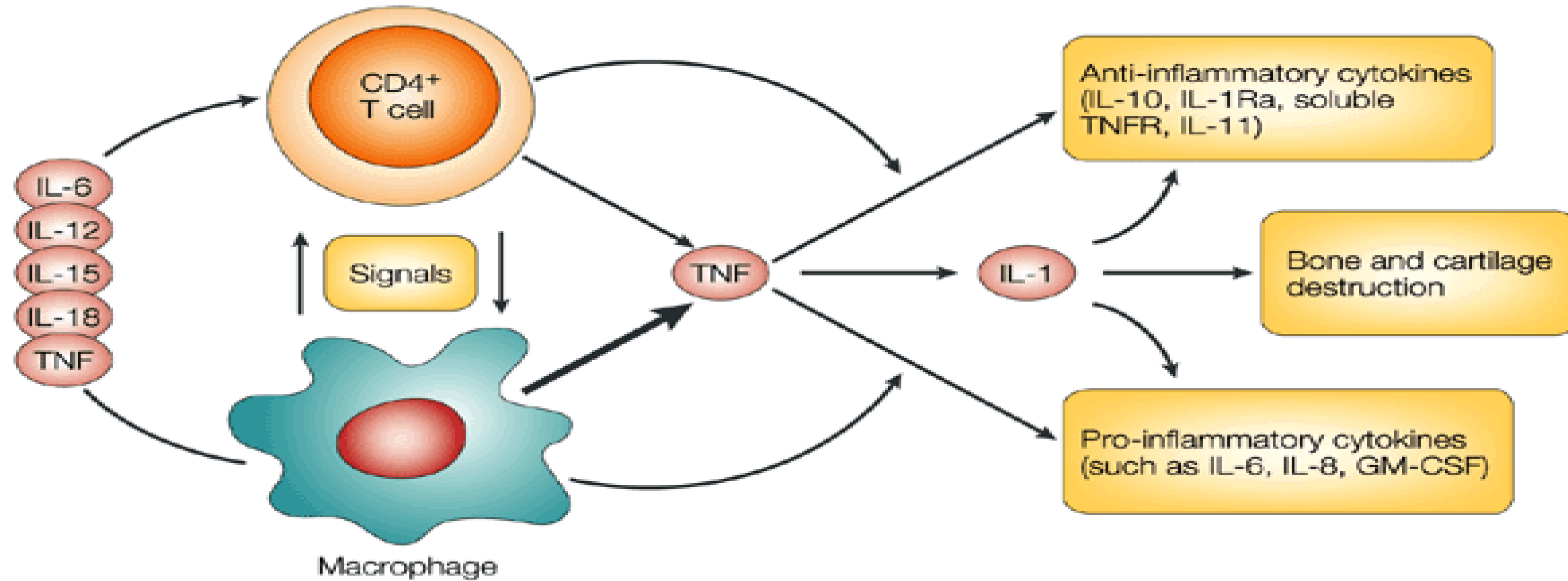
Green regions, polyethylene glycol (PEG)



Source: Bertram G. Katzung, Anthony J. Trevor: Basic & Clinical Pharmacology, 13th Ed.
www.accesspharmacy.com
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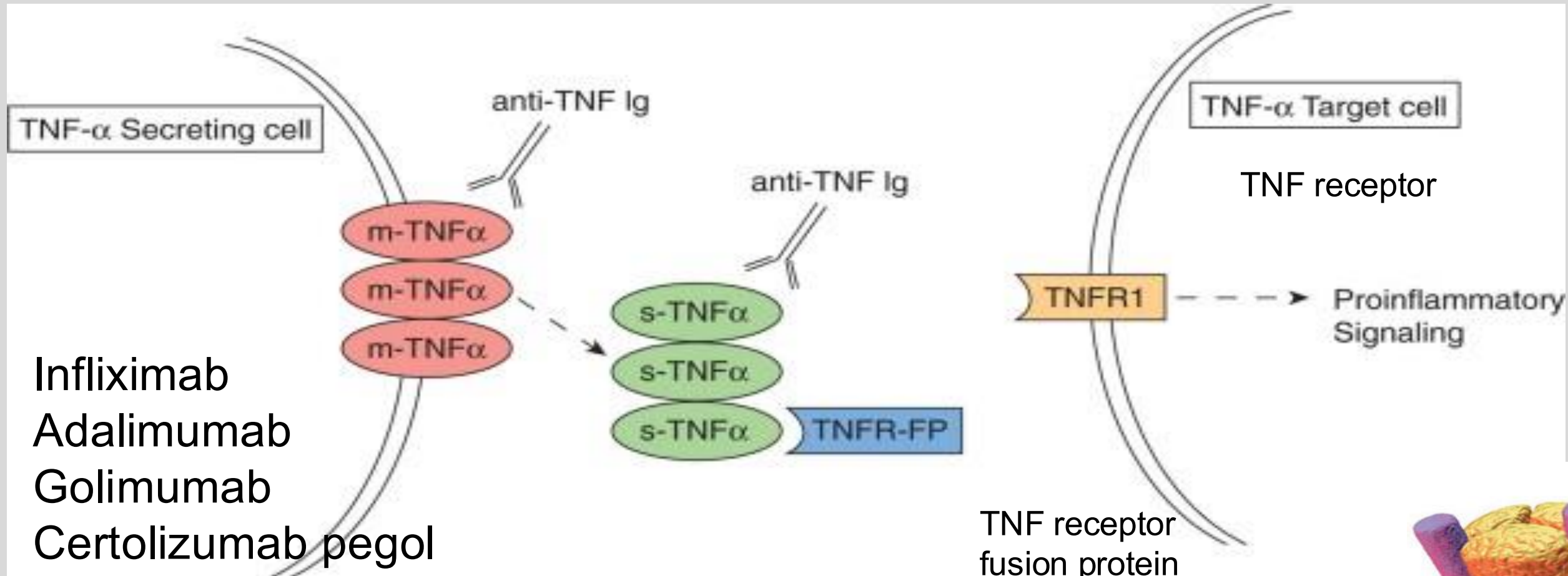
TNF α plays an important role in pathophysiology of in RA.

- TNF α is found in RA synovial fluid and in large concentrations in the serum of RA patients
- Produced by mainly by monocytes and macrophages; also by B cells, T cells and fibroblasts
- Promotes expression and release of cytokines IL-1, IL-6, IL-8, growth factors, adhesion molecules, and others



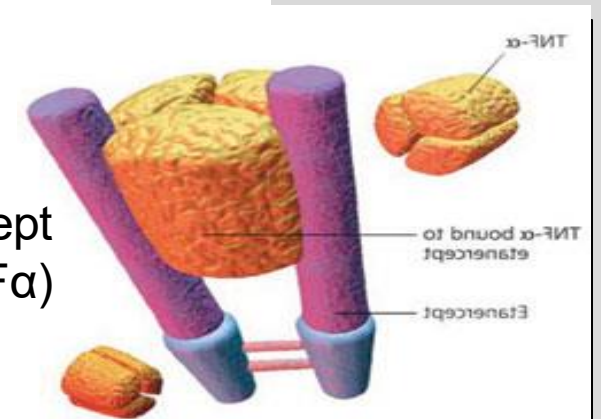
- Some of the cytokine and cellular interactions in the TNF-dependent cytokine cascade to illustrate the crucial role of TNF in the cytokine network of rheumatoid arthritis
- Key interacting cells include CD4⁺ T cells and macrophages, although the signals between them are not fully understood

TNF α inhibitors bind membrane-bound and/or soluble TNF α and neutralize its activity.



TNF receptor
fusion protein

Etanercept
(TNF-FP that binds soluble TNF α)



Infliximab Chimeric Monoclonal Antibody

PK	• IV; maintenance dose every 8 weeks • Elimination undefined • (possibly via mononuclear phagocyte system) • $t_{1/2}$ 7-12 days • Onset of response in RA 3-7 days • Duration of action in RA 6-12 weeks
MOA	• Binds to both soluble and membrane-bound TNFα preventing the cytokine from binding to its receptor
Use	• Rheumatoid arthritis, psoriatic arthritis, active ankylosing spondylitis, Crohn's disease, plaque psoriasis, JIA in children ≥ 4yo • (also used in other inflammatory diseases affecting children)
AEs	• Typical of class; hepatic reactions; infection risk; HBV / TB reactivation; local injection site reactions; hypersensitivity reactions

Etanercept Fusion Protein

PK	SubQ 1-2x weekly (for RA) t _{1/2} adults ~100 hours; t _{1/2} children ~80 hours Onset of effect in RA 2-3 weeks Full effect usually in 3 months
MOA	Binds to soluble TNF α preventing the cytokine from binding to its receptor (it also inhibits lymphotoxin- α , a proinflammatory cytokine)
Use	Rheumatoid arthritis, psoriatic arthritis, active ankylosing spondylitis, plaque psoriasis, JIA in children $\geq$ 2yo
AEs	Typical of class; infection risk; HBV / TB reactivation; local injection site reactions; hypersensitivity reactions (generally well tolerated)

Other Anti-TNF Agents that bind and neutralize soluble and membrane-bound $\text{TNF}\alpha$

Adalimumab	Fully human mAb subQ weekly or every other week $t_{1/2}$ 10-20 days max effect 3 mon; benefit persists ~1 month
Golimumab	Fully human mAb subQ 1x monthly $t_{1/2}$ ~2 weeks
Certolizumab pegol PEG: polyethylene glycol Pegylation increases the $t_{1/2}$.	Humanized Fab fragment of $\text{TNF}\alpha$ has been pegylated subQ; $t_{1/2}$ ~2 weeks It lacks the Fc portion → does not induce complement activation, cell-mediated cytotoxicity, or apoptosis

IL-6R Antagonists

*Tocilizumab

- Sarulimab



Cytokines of the IL-6 superfamily regulate numerous biological processes. They bind to the common signaling receptor membrane-protein subunit gp130 of the IL-6R complex.

Stressors, such as tissue injury or infection, stimulate the production of IL-6 by innate immune cells, such as macrophages and monocytes. Many other cell types also produce IL-6, such as fibroblasts, synovial, and vascular endothelial cells. It is involved in a variety of physiologic processes such as stimulating antibody production, inducing naïve CD4⁺ T cell differentiation into effector cells, monocyte differentiation into macrophages and dendritic cells, activation of vascular endothelial cells that secrete chemokines and adhesion molecules, and many other functions.

Tocilizumab: Humanized MAb IL-6R antagonist

PK	I.V. q4 weeks; subQ weekly or every other week depending on body weight, $t_{1/2\text{terminal}}$ up to 13 days at steady state
MOA	IL-6 receptor antagonist Binds to soluble and membrane-bound receptors Inhibition of IL-6 receptor leads to reduction in cytokine production
Use	RA treatment as monotherapy or combined with methotrexate Adults and JIA in children $\geq 2\text{yo}$ COVID-19, hospitalized patients receiving supplemental oxygen and mechanical ventilation (EUA) EUA: emergency use authorization
AE	Infection risk, neutropenia/thrombocytopenia, $\uparrow$ liver enzymes, $\uparrow$ LDL -C; GI perforation (reported, typically 2 nd to diverticulitis)
Pregnancy	Adverse effects observed in animal reproduction studies

bDMARDS: agents that suppress T cell and B cell proliferation and function

* Abatacept

* Rituximab

T cells:

Antigen presentation
+ costimulation of
key cell surface receptors



T cell activation and
differentiation



release of
pro-inflammatory cytokines



instigates inflammatory cascade

B cells:

Activated T cells activate B cells



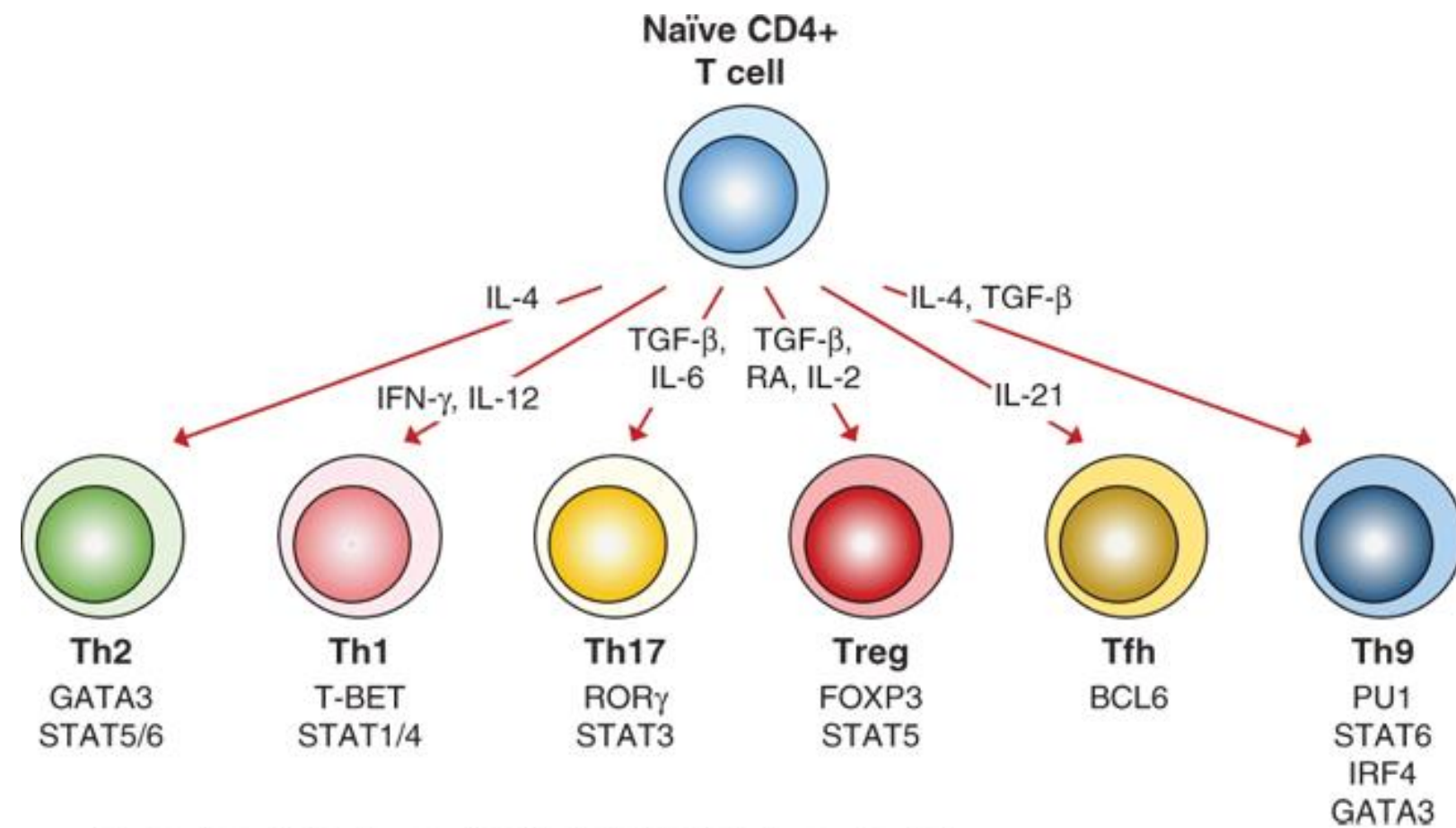
differentiation: memory and
plasma cells → antibody
production



antigen-antibody complex
formation

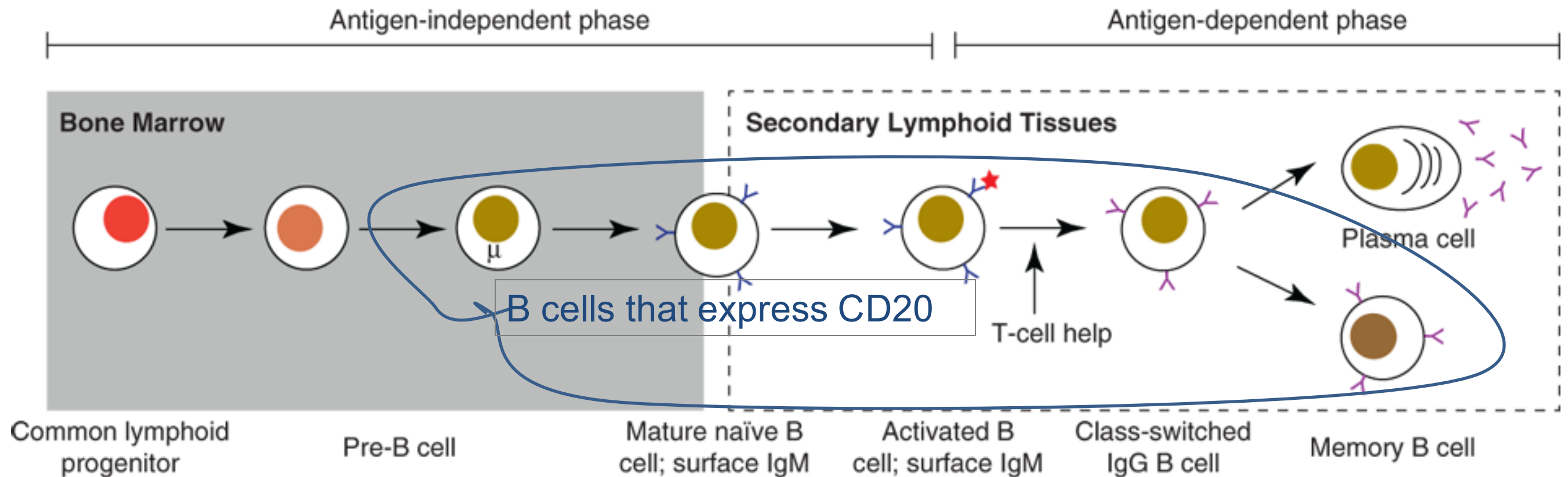


complement activation



Source: Kenneth Kaushansky, Josef T. Prchal, Linda J. Burns, Marshall A. Lichtman, Marcel Levi, David C. Linch: *Williams Hematology*, 10e
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Differentiation of CD4⁺ T-cell subsets. During the course of the immune response, a naïve CD4⁺ T cell (top of figure) can differentiate into any one of several distinctive CD4⁺ T cells, as indicated beneath each differentiated cell type. Beneath the name of each T-cell subset is listed the transcription factor (if known) that is critical for the differentiation and maintenance of the subset. Interleukin (IL)-4 is critical for the development of Th2 cells (green), which triggers activation and/or induction of signal transducer and activator of transcription (STAT) 5, STAT6, and GATA3 (transacting T-cell-specific transcription factor), transcription factors that are important in Th2 differentiation. On the other hand, interferon (IFN)- γ and IL-12 pattern the development of Th1 cells (light red) through the activation and/or induction of transcription factors STAT1, STAT4, and T-bet. Interleukin (IL)-23 along with IL-1 β or transforming growth factor β (TGF- β) can induce blood CD4⁺ T cells to express the transcription factors STAT3 and ROR γ t (retinoic acid-related orphan receptor γ thymus isoform), which programs differentiation into Th17 cells (light yellow), whereas TGF- β , retinoic acid (RA), and IL-2 induce these cells to express the transcription factors FOXP3 (forkhead box P3) and STAT5, which are required for differentiation of these cells into T regulatory (Treg) cells (red). Finally, IL-21 favors differentiation of naïve CD4 T cells into follicular helper T cells (Tfhs [gold]). There is some plasticity in these differentiated T-cell subsets. IL-6 in combination may induce Treg cells to differentiate into Th17 cells, whereas B cells stimulated via CD40-CD40-ligand (CD40L or CD154) may induce their differentiation into Tfh cells, as indicated by the dashed horizontal arrows. Similarly, IL-21 may induce Th17 cells to differentiate into Th1 cells, whereas IL-4 may induce their differentiation into Th2 cells. Th9 cells (dark blue) constitute a putative CD4⁺ T-cell subset characterized by its ability to secrete IL-9. These cells apparently differentiate from Th2 cells in the context of TGF- β . BCL, B-cell lymphoma protein 6; IRF, interferon regulatory transcription factor; PU1, transcription factor binding to the PU box.

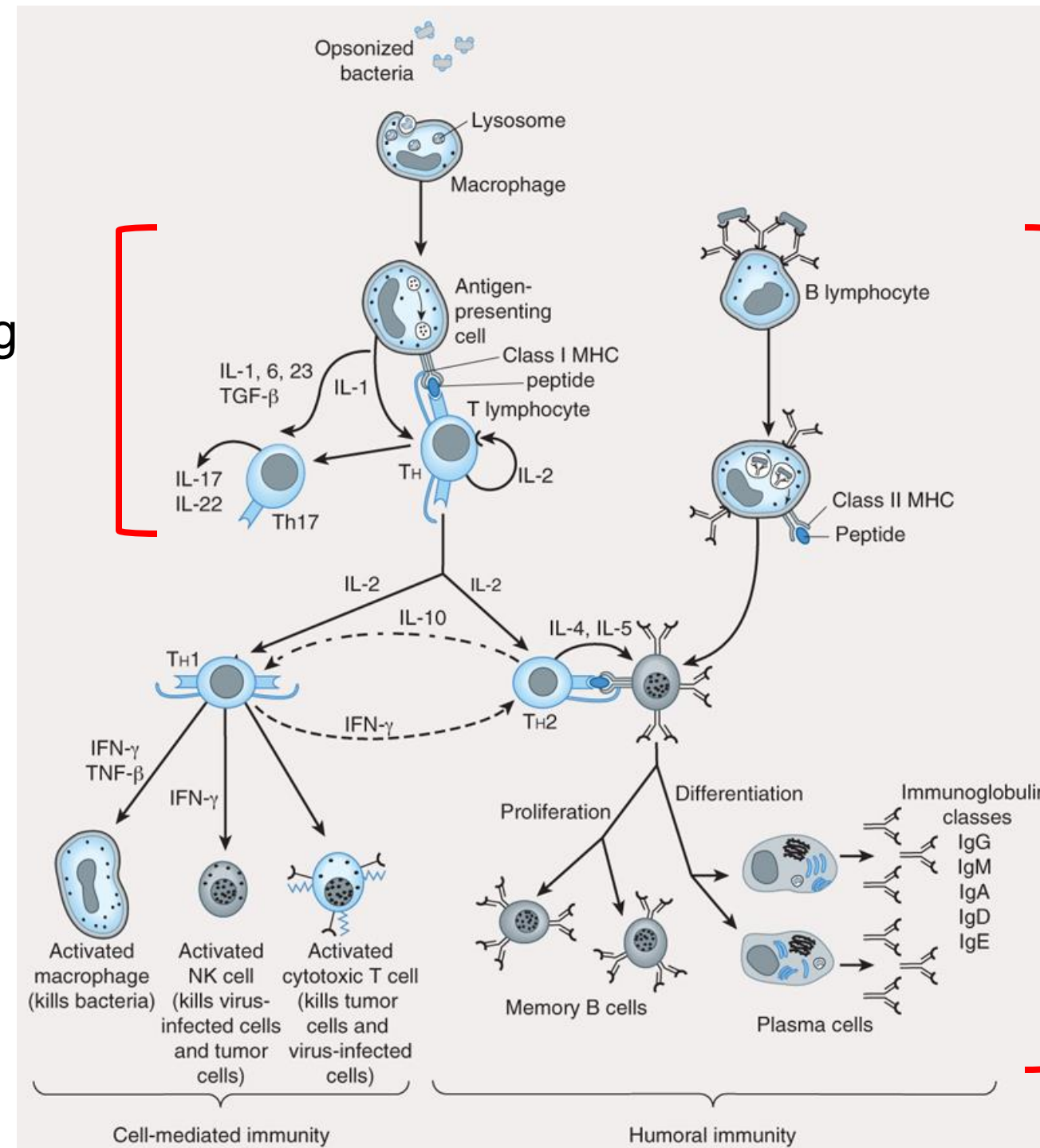


Source: W. Levinson, P. Chin-Hong, E.A. Joyce, J. Nussbaum, B. Schwartz:
 Review of Medical Microbiology & Immunology: A Guide to Clinical Infectious
 Diseases, Seventeenth Edition: Copyright © McGraw Hill. All rights reserved.

CD20 is a cell surface antigen expressed on the surface of all B cells beginning with the pro-B cell stage through its terminal differentiation to plasma cells. Anti-CD20 therapeutics were originally used to treat B cell cancers and are now used in the management of many autoimmune diseases.

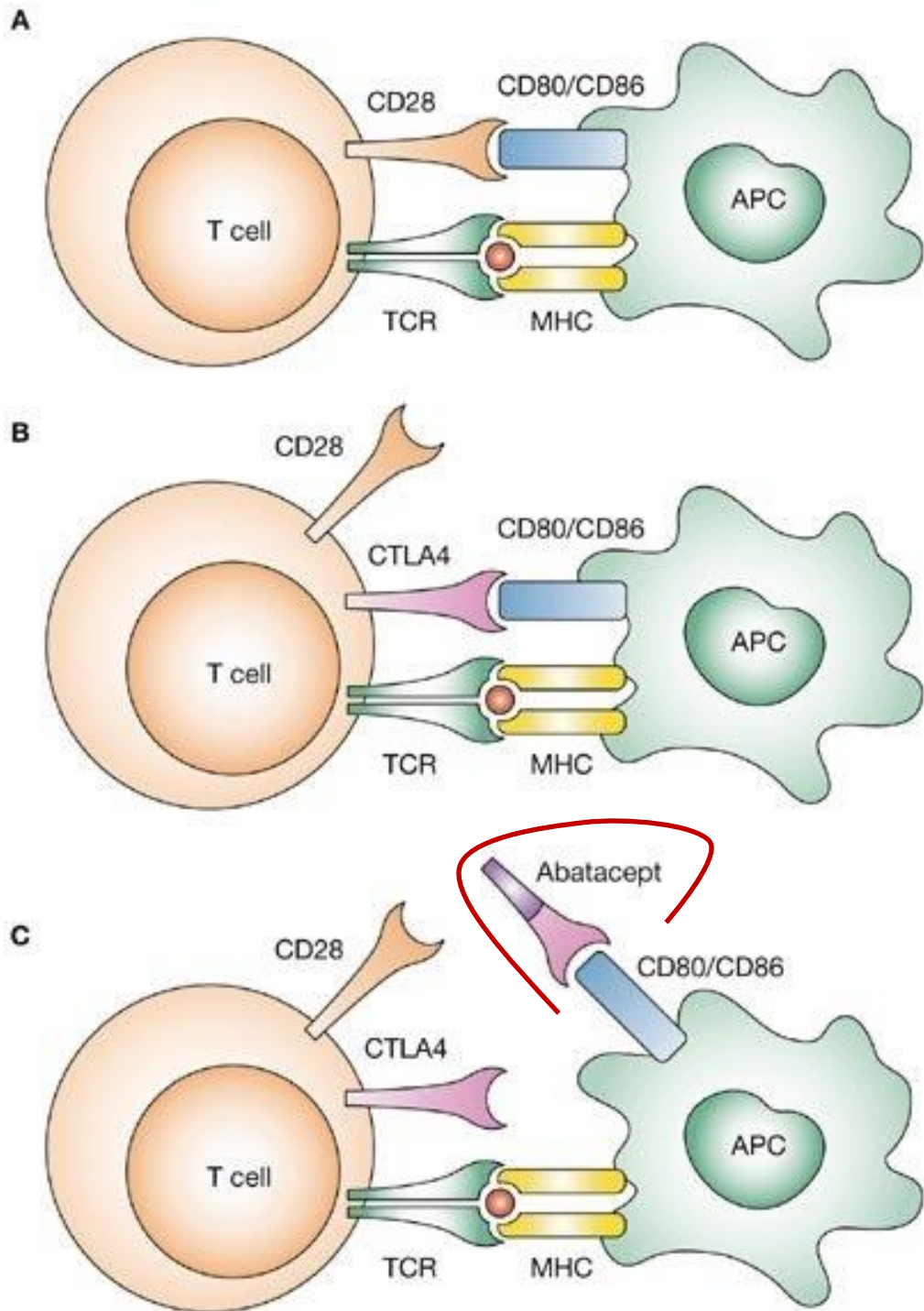
Maturation of B cells. B cells arise from lymphoid progenitor stem cells and differentiate into pre-B cells expressing μ heavy chains in the cytoplasm and then into mature B cells expressing monomer IgM on the surface. This occurs independent of antigen. Activation of B cells, class switching, and differentiation into memory B cells and plasma cells occurs after exposure to antigen (red star) and is enhanced by T-cell help. μ = mu heavy chains in cytoplasm; Y = IgM (blue) or IgG (purple). Adapted with permission from Stites DP, Terr A: Basic & Clinical Immunology, 7th ed. New York, NY: McGraw Hill; 1991.

Presentation of peptide fragment to T cell, leading to T cell activation and triggering an immediate immune response.



T_H2 cells activate memory B cells, leading to proliferation of memory B cells and differentiation into plasma cells. B cells are also professional antigen presenting cells (MHC class II).

Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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Abatacept Mechanism

Fusion Protein: CTLA-4 fused to IgG

1. Antigen presentation: APC presents antigen on MHC complex to T cell receptor complex
2. Costimulatory signal: CD80/86 on APC interacts with CD28 on T cell → T cell activation → clonal expansion
3. Inhibition: Activated T cell expresses CTLA-4; CD80/CD86 binds CTLA-4
 - CD80/CD86 has greater affinity for CTLA-4 than for CD28
4. T cell activity is inhibited
 - **Abatacept binds CD80/86 on APC and blocks costimulatory signal.**

APC: antigen presenting cell; TCR: T cell receptor; CTLA: cytotoxic T lymphocyte antigen 4; Note: CD80/86=B7.1/B7.2

Abatacept Fusion Protein: CTLA-4 fused to IgG

PK	I.V. q4 weeks (maintenance); subQ once weekly t _{1/2} 8-25 days
MOA	Binds CD80/86 on APC → blocks costimulatory interaction between APC and CD28 on the T cell → blocks costimulatory signal → inhibits T cell activation (see previous slide)
Use	RA with or without MTX; JIA; psoriatic arthritis; graft-vs-host disease (GVHD) prophylaxis
AE	Serious infections, respiratory infections, local injection site reactions possible; overall safety comparable to TNF inhibitors
Caution	Coadministration with other targeted immunosuppressants is contraindicated

Rituximab Mechanism

Anti-CD20 MAb

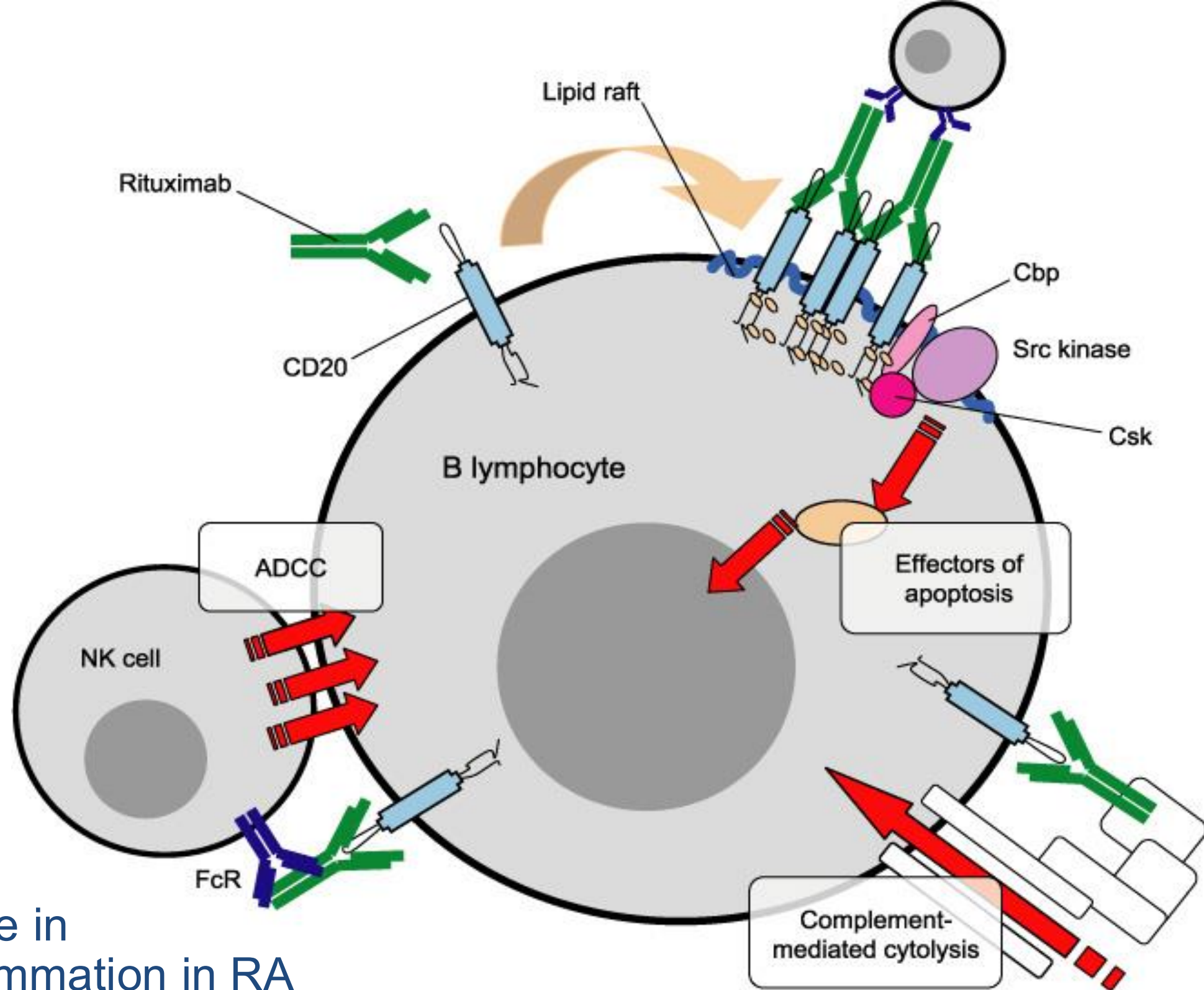
Rituximab induces depletion of peripheral B cells by:

- Antibody-dependent cellular cytotoxicity (ADCC)
- Complement-dependent tumor cell lysis
- Apoptosis by direct binding to and cross-linking CD20

↓ antigen presentation to T cells

↓ inflammation

CD20+ B cells play a critical role in perpetuating synovitis and inflammation in RA



Rituximab: Chimeric MAb against CD20+ B cells

PK	I.V. infusion q24 weeks for RA (maintenance regimen) mean $t_{1/2\text{terminal}}$ 18 days (range in RA 5-78 days)
MOA	Depletes CD20+ B cells (previous slide) Antibody-dependent cellular cytotoxicity (ADCC) Complement-dependent tumor cell lysis Apoptosis (direct binding to and cross-linking CD20)
Use	RA – inadequate response to MTX and anti-TNF therapy, or a history of lymphoproliferative disorder treatable by rituximab CD20+ B cell lymphomas / chronic lymphocytic leukemia GvHD; several autoimmune diseases (numerous on- and off-label uses)
AE	Infusion/hypersensitivity reactions; rash; hypo- or hypertension; arrhythmia; reactivation of HBV infection; PML
Caution	Avoid in pregnancy; not approved for use in children with JIA

HBV: hepatitis B virus; PML: progressive multifocal leukoencephalopathy (JC virus)

Recap: bDMARDs

bDMARDS			
TNF Inhibitors: Infliximab Adalimumab Etanercept Golimumab Certolizumab pegol	Neutropenia Infections Reactivation of latent TB Lymphoma risk Demyelinating disease Heart failure Autoimmunity induction	Infusion reaction Neutralizing antibodies/loss of efficacy	PPD skin test Viral hepatitis screening LFTs periodically Signs of infection, heart failure, hypersensitivity reactions, lupus-like syndrome, malignancy
Tocilizumab IL-6R inhibitor	Serious infections Neutropenia Thrombocytopenia Hypersensitivity reactions	LFT elevation Dyslipidemia Constipation Infusion reactions	PPD skin test CBC, LFTs, Lipid panel Signs of infection CNS demyelinating disorder GI perforation Hypersensitivity reactions
Sarilumab IL-6R inhibitor			
Abatacept CTLA-Ig binds CD80/86	Serious infections Malignancy Hypersensitivity reactions	Hypertension Headache Nausea Neutralizing antibodies/loss of efficacy Respiratory tract infection	PPD skin test Signs of infusion reactions and Infection Periodic skin examinations for skin cancer
Rituximab CD20 inhibitor	Infections Infusion reactions Mucocutaneous reactions Cytopenia Cardiac disorder PML (case reports)	Cardiac disorder Rash Fever Neutralizing antibodies/loss of efficacy	Viral hepatitis screening CBC at regular intervals Preexisting heart disease Signs of infection, CNS demyelinating disorder, cardiac disorder, mucocutaneous reactions

Oral small molecule targeted DMARDs (tsDMARDs)

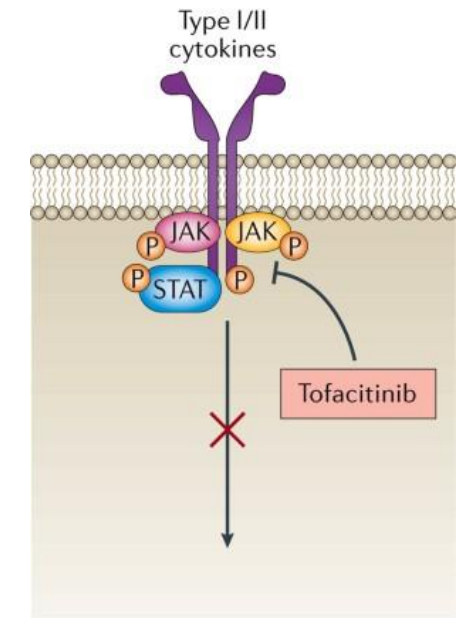
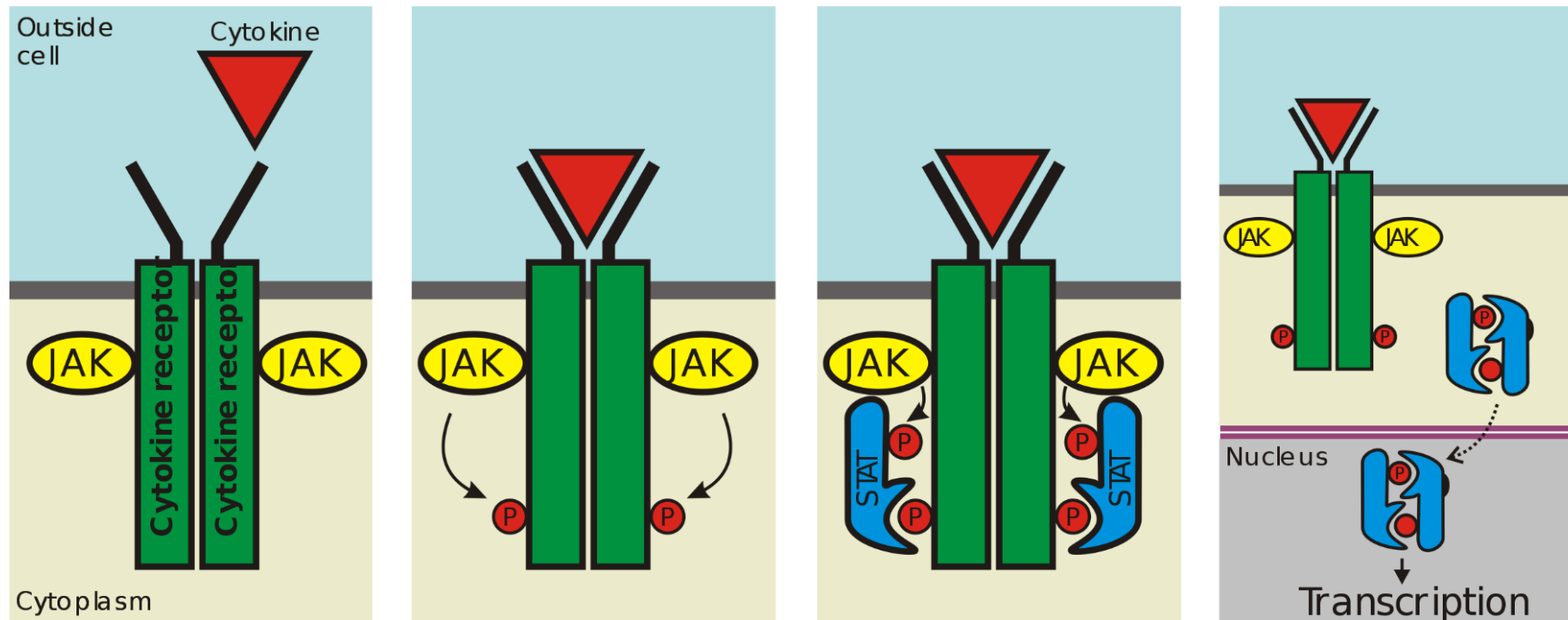
JAK Inhibitors

*Tofacitinib (inhibits JAK1, JAK3, others)

- Baricitinib (preferentially inhibits JAK1, JAK2)
- Upadacitinib (preferentially inhibits JAK1)

JAK inhibition prevents signal transduction for numerous cytokines, which inhibits transcription of several genes involved in differentiation, proliferation, and function of NK cells, T cells, and B cells.

JAKs are intracellular enzymes involved in stimulating hematopoiesis and immune cell function.



Nature Reviews | Rheumatology

JAK: Janus kinase, an intracellular tyrosine kinase; STAT: Signal transducers and activators of transcription

1. Ligand (interleukin, interferon) binding → receptor homo- or heterodimerization bringing JAKs close together.
2. JAKs phosphorylate each other (transphosphorylation) → JAK activation.
3. JAKs phosphorylate sites on the receptors → binding sites for other proteins.
4. STATs then bind. They are phosphorylated by JAKs.
5. The STATs dissociate and homo- or heterodimerize and translocate to nucleus → induce transcription of target genes.

Tofacitinib: JAK inhibition interrupts signaling of several cytokines.

- Oral
- Moderate protein binding
- Hepatic metabolism via CYP3A4 mainly
- Renal excretion (30% unchanged)
- $t_{1/2\text{elimination}} \sim 3$ hours

RA / Chronic inflammatory diseases:

- Use limited to patients who have not adequately responded to or are intolerant of TNF inhibitors
- COVID-19, alternative to baricitinib in hospitalized pts on supplemental O₂, mechanical ventilation, and glucocorticoid $\pm$ remdesivir

- Thrombosis / CV adverse events (MI, stroke, VTE, pulmonary embolism)
- Higher rate all-cause mortality, pts ≥ 50 yo with ≥ 1 CV risk factor
- Lymphopenia, neutropenia, anemia
- Malignancy (lymphoma, lung cancer, etc.)
- Infection
- Gastrointestinal perforation, esp. pts. with diverticulitis
- Avoid in pregnancy
- bDMARDs combo contraindicated
- CYP3A4-mediated drug interactions and potent immunosuppressants (contraindicated)

CYTOKINE INHIBITORS

for Treatment of Psoriasis, Psoriatic Arthritis, and other Inflammatory Diseases

Drug target Drug name	Potentially serious toxicities	Common adverse effects	Screening, Baseline Tests, and Monitoring
IL-1Ra inhibitor Anakinra	Serious infections Neutropenia	Respiratory infections Nasopharyngitis	PPD skin test CBC with diff
IL-1 β inhibitor Canakinumab Rilonacept	Hypersensitivity reactions	Musculoskeletal pain Diarrhea	C-reactive protein Acute phase SAA
IL-17 inhibitor Secukinumab	- Inflammatory bowel disease new onset or exacerbation - Serious infections - Hypersensitivity reactions - Reactivation of latent infections	- Upper respiratory tract infection - Neutropenia - Injection site reaction	PPD skin test Viral hepatitis screening HIV screening (high risk patients) CBC with diff Complete metabolic panel
IL-17A inhibitor Ixekizumab			
IL-12/23 Inhibitor Ustekinumab IL23, B19 subunit inhibitor Guselkinumab Risankizumab Tildakizumab	Infection; malignancy (cutaneous squamous cell carcinomas); Reversible posterior leukoencephalopathy Hypersensitivity reactions	Infection Nasopharyngitis Headache Fatigue	PPD skin test Viral hepatitis screening CBC with diff Signs/symptoms of infection, posterior leukoencephalopathy, squamous cell skin carcinoma

Check your knowledge

Annabelle is 52-year-old woman with RA diagnosed 1 year ago. Her current RA regimen consists of methotrexate at maximal dose, infliximab, and a prednisone because of recurrent disease flares. Annabelle agrees to discontinue the infliximab and begin treatment with a cytokine receptor antagonist that blocks the actions of a cytokine found in high concentrations in the synovium and serum of RA patients. She will receive intravenous induction therapy with the biologic drug followed by maintenance biweekly self-administered subQ injections. Monitoring includes complete blood count, liver function tests, and signs/symptoms of infection.

What is the drug?

- Abatacept
 - Etanercept
 - Rituximab
 - Tocilizumab
 - Tofacitinib
-

Summary
csDMARDs
bDMARDs
tsDMARDs

- Disease-modifying antirheumatic drugs (DMARDs) are a heterogeneous group of drugs that have anti-inflammatory actions.
- Early, aggressive treatment of RA slows the progression of joint damage by modifying the underlying disease. It may take several weeks to months for the antirheumatic effects of DMARDs to become apparent.
- DMARDs are sub-categorized by their structures into the conventional synthetic DMARDs, biologic DMARDs, and targeted small molecule synthetic DMARDs.
- The csDMARDs currently used are methotrexate, hydroxychloroquine, sulfasalazine, and leflunomide.

The bDMARDs and tsDMARDS are often used in combination with MTX for enhanced efficacy.

bDMARDs

TNF α inhibitors (first-line)	Infliximab Etanercept	Adalimumab Golimumab	Certolizumab pegol
IL-6 receptor inhibitor	Tocilizumab	Sarilumab	
T cell costimulation inhibitor	Abatacept		
B-cell cytotoxic agent	Rituximab		

tsDMARDs

JAK inhibitors	Tofacitinib	Baricitinib	Upadacitinib
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- Each drug has specific pharmacokinetic properties and mechanisms of anti-inflammatory and antiproliferative actions.
- Combinations of csDMARDs or combinations of MTX or other csDMARD with a bDMARD or tsDMARD are used in patients who have had an inadequate response to anti-TNF α agents.
- Such combinations have shown improved efficacy.
- Therapy must be tailored to the individual patient.
- Adverse effects are related to their powerful immunosuppressive effects and their proteinaceous nature: reactivation of latent infection, risk of serious infection and malignancy; injection reactions, hypersensitivity reactions; neutralizing antibodies to the biological agents may reduce efficacy.
- Patients should be screened for exposure to TB, HBV, and HCV.
- Concomitant strong immunosuppressive drugs and drugs having similar toxicities or metabolism/transport-mediated drug-drug interactions can increase the risk of adverse drug events.

Introductory Case Vignette

HINT: Connect the monitoring tests to the adverse effects and the patient’s renal function.

- Justine is a 45-year-old female diagnosed with moderately to severely active RA. The rheumatologist discusses early aggressive therapy with her.
- Justine agrees to oral triple therapy with glucocorticoid bridging.

What medications will Justine be receiving for RA treatment?

What monitoring for adverse effects should be done relative to each drug?

Prednisone high dose with tapering	HBV, HCV, TB testing; BP, serum glucose, electrolytes, weight, presence of infection
Methotrexate once weekly with daily folic acid supplement	HBV, HCV, TB testing; CBC with differential Renal function tests; Liver function tests Signs/Symptoms of stomatitis
Sulfasalazine	CBC with differential Renal function tests; Liver function tests Signs/Symptoms of skin rash
Hydroxychloroquine	Ophthalmologic exam at baseline and annually CBC with diff, Renal function and Liver function tests EKG in patients at risk for QT interval prolongation Signs/Symptoms of hypoglycemia, muscle weakness

Check your knowledge: Answers

June is a 28-year-old woman given a diagnosis of moderate RA without features of a poor prognosis. She was recently married and reports that she and her husband are planning to start a family.

Which of the following would be an appropriate initial RA monotherapy for this patient?

- Hydroxychloroquine
- Leflunomide
- Methotrexate
- Sulfasalazine

Answer: Sulfasalazine

MTX and LEF are contraindicated in pregnancy.

SSZ monotherapy in RA is more effective than HCQ for moderate RA.

HCQ is a reasonable alternative for patients with mild RA.

SJ is 45-year-old man who presents to the ED with complaints of nausea and vomiting, lack of appetite, right upper quadrant pain, abdominal distention, generalized itching, jaundice, fatigue, and confusion. Medical history is significant for moderate to severe RA treatment with an oral csDMARD. Right upper quadrant tenderness, hepatomegaly, and ascites are noted on PE.

Prothrombin time is prolonged and aminotransferase and bilirubin levels are elevated. The patient is admitted for management of drug-induced acute liver failure. Cholestyramine is included in the treatment to accelerate drug elimination by preventing enterohepatic circulation of the RA drug.

- What is the drug? Answer: Leflunomide ($t_{1/2}$ ~15 days) is associated with acute liver toxicity.

Back to Mary's case:

Mary presents to the rheumatologist for a 3-month evaluation of her rheumatoid arthritis treatment.

- In the past month, Mary experienced a disease flare treated with a 3-week course of high-dose prednisone followed by dose tapering and withdrawal of the glucocorticoid. She is tolerating the methotrexate.
- As monotherapy with MTX alone was inadequate for controlling disease activity, Mary is offered oral triple therapy or MTX / bDMARD therapy.
- Mary opts for the biologic agent because therapeutic response is achieved more quickly than with the triple therapy. She refuses the intravenous bDMARD option and agrees to add-on therapy with adalimumab self-injected subcutaneously every other week, which is covered by her insurance plan.

Adalimumab is in what class of drugs (mechanism-based classes)? **TNF inhibitor**

What is the intravenous option that Mary preferred not to receive? **Infliximab**

What are other therapy options for the treatment of RA resistant to methotrexate monotherapy?

Leflunomide alternative to MTX or with MTX (effective combo but higher incidence of toxicity)

MTX plus abatacept, tocilizumab / sarilumab, or tofacitinib / baricitinib / upadacitinib

Check your knowledge: Answer

Annabelle is 52-year-old woman with RA diagnosed 1 year ago. Her current RA regimen consists of methotrexate at maximal dose, infliximab, and prednisone because of recurrent disease flares. Annabelle agrees to discontinue the infliximab and begin treatment with a cytokine receptor antagonist that blocks the actions of a cytokine found in high concentrations in the synovium and serum of RA patients. She will receive intravenous induction therapy with the biologic drug followed by maintenance biweekly self-administered subQ injections. Monitoring includes complete blood count, liver function tests, and signs/symptoms of infection.

What is the drug?

- Abatacept
- Etanercept
- Rituximab
- Tocilizumab
- Tofacitinib

Answer: Tocilizumab – the only cytokine receptor antagonist in the options
It binds soluble and membrane-bound IL-6 receptors. IL-6 mediates a variety of immunological responses.

Inhibition of IL-6 receptor reduces cytokine production.

- Abatacept: CTLA-Ig binds CD80/86, which prevents T cell costimulation
- Etanercept: Binds and neutralizes soluble TNF-alpha
- Rituximab: Binds CD20 on B cells inducing depletion of peripheral B cells
- Tofacitinib: Oral JAK inhibitor; prevents activation of STATs gene expression and intracellular activity of immune cells.

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“Doctor and Doll”
Norman Rockwell (1929)

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